

Partial Progress on Exact-Nash Convergence in Finite-State Ergodic Mean Field Games: A Contractive Stationary Regime

Abstract

This note gives a rigorous conditional convergence theorem for exact Nash equilibria in finite-state, infinite-horizon ergodic Markovian games. It is *partial progress*, not a solution of the general convergence problem posed in Remark 4.6 of Cohen and Zell. We consider symmetric stationary anonymous exact equilibria of the form

$$\alpha^{n,i}(\mathbf{x}) = A_{x_i}^n(M^{-i,n}(\mathbf{x})).$$

We assume that the feedback maps A^n are uniformly Lipschitz in the empirical measure and that the induced nonlinear population drifts $b^n(m) = mA^n(m)$ are uniformly strongly contractive. Under these assumptions, we prove a uniform Lyapunov estimate that remains valid after an arbitrary unilateral Markovian deviation. Exact Nash optimality then implies that the deterministic invariant pair selected by the finite-player equilibrium has stationary mean field optimality gap $O(n^{-1/2})$. Every subsequential limit is therefore a stationary mean field equilibrium. If the stationary mean field equilibrium measure is unique, the invariant empirical laws, all fixed player marginals, and the exact equilibrium values converge; if the equilibrium generator is also unique, the equilibrium controls converge in stationary mean. The unresolved part of the original problem is to derive the required uniform feedback regularity and stability from the finite-player Nash system, and to treat arbitrary time-dependent or nonanonymous exact Markovian equilibria.

Status relative to the open problem. The theorem below is complete for the stated stationary, symmetric, anonymous, uniformly regular, contractive class. It should nevertheless be classified as *partial progress* on the problem in [1, Remark 4.6], because the central uniform regularity of the exact-equilibrium feedbacks is assumed rather than derived from the Nash system, and the theorem does not cover general Markovian exact equilibria.

1 Introduction

Cohen and Zell construct approximate Nash equilibria of finite-state ergodic n -player games from solutions of the mean field game system and from the ergodic master equation. They explicitly leave open the reverse direction: whether exact Markovian Nash equilibria of the finite-player games converge to a mean field equilibrium as $n \rightarrow \infty$; see [1, Remark 4.6]. The difficulty is not merely a finite-horizon propagation-of-chaos estimate. In a closed-loop game, a unilateral deviation changes the deviator's state, the other players observe that state through their empirical arguments, and their controls react. An $O(1/n)$ perturbation can therefore feed back into the population over an arbitrarily long horizon.

This note isolates a regime in which that amplification cannot occur. We assume that a sequence of exact equilibria is stationary, symmetric, and anonymous, with a common feedback map $A^n(m)$. Two uniform assumptions are imposed:

- (i) A^n is Lipschitz in the empirical measure, uniformly in n ;
- (ii) the induced population drift $b^n(m) = mA^n(m)$ satisfies a uniform strong one-sided contraction inequality.

The first assumption makes the direct effect of one player on the opponents' rates of order $1/n$. The second prevents this perturbation from being amplified over the infinite horizon.

The main argument has three steps. First, contraction gives a unique population fixed point m^n and an $O(1/n)$ stationary mean-square concentration estimate for the empirical measure. Second, the same Lyapunov estimate holds for the empirical measure of the $n - 1$ nondeviating players, uniformly over the deviator's strategy. Third, testing the exact Nash inequality against stationary own-state deviations converts finite-player optimality into the continuum residual

$$\mathcal{G}(m^n, A^n(m^n)) = O(n^{-1/2}),$$

where $\mathcal{G}$ is the frozen stationary mean field optimality gap. Compactness then identifies every subsequential limit as a stationary mean field equilibrium.

The result is directly relevant to the open problem, but it does not settle it. In particular, it does not prove a uniform empirical Lipschitz estimate for solutions of the finite-player Nash system, does not prove contraction of all exact-equilibrium drifts from the assumptions in [1], and does not treat arbitrary time-dependent or nonanonymous exact equilibria. The precise scope is summarized again in [section 9](#).

2 Model and stationary mean field equilibria

2.1 Rate matrices and costs

Let

$$\mathcal{S} = \{1, \dots, d\}, \quad \mathcal{P} = \left\{ m \in [0, 1]^d : \sum_{x=1}^d m_x = 1 \right\}.$$

Probability vectors are row vectors. We use the Euclidean norm and inner product for vectors unless stated otherwise; matrix norms are Frobenius norms unless an operator-norm subscript is displayed. Product spaces carry the corresponding Euclidean product norm, and W_2 denotes the Wasserstein metric induced by the Euclidean distance on $\mathcal{P}$. Fix constants $0 < \underline{a} \leq \bar{a} < \infty$. For $x \in \mathcal{S}$, let

$$\mathcal{A}_x := \left\{ a \in \mathbb{R}^d : a_y \in [\underline{a}, \bar{a}] \text{ for } y \neq x, \quad a_x = - \sum_{y \neq x} a_y \right\}.$$

Let $\mathcal{A}$ denote the compact set of $d \times d$ rate matrices A whose x th row A_x belongs to $\mathcal{A}_x$. Every $A \in \mathcal{A}$ is irreducible, so it has a unique invariant probability vector $\pi(A)$.

The running cost is

$$c(x, a, m) = f(x, a) + F(x, m).$$

Assumption 2.1 (Cost regularity). *There are constants $L_f, L_F < \infty$ such that, uniformly in x ,*

$$|f(x, a) - f(x, a')| \leq L_f \|a - a'\|, \quad |F(x, m) - F(x, m')| \leq L_F \|m - m'\|.$$

In particular, c is bounded on its compact domain.

2.2 The frozen ergodic control problem

For $m \in \mathcal{P}$ and $A \in \mathcal{A}$, define

$$\mathcal{J}(A; m) := \sum_{x \in \mathcal{S}} \pi_x(A) c(x, A_x, m), \quad \mathcal{V}(m) := \min_{A \in \mathcal{A}} \mathcal{J}(A; m). \quad (1)$$

The next standard fact explains why minimizing over stationary rate matrices is sufficient for the frozen one-player average-cost problem.

Lemma 2.2 (Stationary controls suffice). *Fix $m \in \mathcal{P}$. Among all admissible, possibly time-dependent Markov controls for a single controlled chain facing the fixed population m , the minimum long-run average cost equals $\mathcal{V}(m)$ and is attained by a deterministic stationary rate matrix $A^m \in \mathcal{A}$.*

Proof. Uniformization reduces this finite-state continuous-time control problem to a finite-state discrete-time average-cost Markov decision process with compact action sets and continuous data. The standard average-cost optimality equation therefore has a solution $(\rho, h) \in \mathbb{R} \times \mathbb{R}^d$:

$$\rho = \min_{a \in \mathcal{A}_x} \{c(x, a, m) + a \cdot h\}, \quad x \in \mathcal{S}, \quad (2)$$

with a measurable minimizing selector A_x^m ; see, for example, [3, Chapters 5–6]. For any admissible Markov control a_t and its controlled state X_t , (2) gives

$$c(X_t, a_t, m) + a_t \cdot h \geq \rho.$$

Dynkin's formula yields

$$\mathbb{E}h(X_T) - h(X_0) = \mathbb{E} \int_0^T a_t \cdot h \, dt.$$

After integration, division by T , and passage to the limit, boundedness of h implies that every admissible control has long-run average cost at least ρ .

For the stationary selector A^m , equality holds in (2). Multiplying by the invariant distribution $\pi(A^m)$ and summing over x gives

$$\rho = \sum_x \pi_x(A^m) c(x, A_x^m, m) = \mathcal{J}(A^m; m),$$

because $\pi(A^m)A^m = 0$. Hence A^m is optimal and $\rho = \mathcal{V}(m)$. $\square$

Definition 2.3 (Stationary mean field equilibrium). *A pair $(m, A) \in \mathcal{P} \times \mathcal{A}$ is a stationary mean field equilibrium if*

$$mA = 0, \quad \mathcal{J}(A; m) = \mathcal{V}(m). \quad (3)$$

Its ergodic value is $\mathcal{V}(m)$.

For every invariant pair $mA = 0$, define the frozen optimality gap

$$\mathcal{G}(m, A) := \mathcal{J}(A; m) - \mathcal{V}(m) \geq 0. \quad (4)$$

Thus $\mathcal{G}(m, A) = 0$ exactly when (m, A) is a stationary mean field equilibrium. Write

$$\mathcal{E} := \{(m, A) \in \mathcal{P} \times \mathcal{A} : mA = 0, \mathcal{G}(m, A) = 0\}. \quad (5)$$

2.3 The finite-player game

For $\mathbf{x} = (x_1, \dots, x_n) \in \mathcal{S}^n$, define

$$M^n(\mathbf{x}) := \frac{1}{n} \sum_{i=1}^n \mathbf{e}_{x_i}, \quad M^{-i,n}(\mathbf{x}) := \frac{1}{n-1} \sum_{j \neq i} \mathbf{e}_{x_j}.$$

An admissible Markovian strategy for player i is a measurable rate row $\beta^i(t, \mathbf{x}) \in \mathcal{A}_{x_i}$. Under a Markovian profile $\boldsymbol{\alpha} = (\alpha^1, \dots, \alpha^n)$, player i has ergodic cost

$$J_i^n(\boldsymbol{\alpha}; \mathbf{x}) := \limsup_{T \rightarrow \infty} \frac{1}{T} \mathbb{E}_{\mathbf{x}} \int_0^T c(X_t^i, \alpha^i(t, \mathbf{X}_t), M_t^{-i,n}) dt. \quad (6)$$

A profile is an exact Markovian Nash equilibrium if, for every player, initial configuration, and admissible Markovian deviation β^i ,

$$J_i^n(\boldsymbol{\alpha}; \mathbf{x}) \leq J_i^n((\boldsymbol{\alpha}^{-i}, \beta^i); \mathbf{x}). \quad (7)$$

We consider the stationary symmetric anonymous class

$$\alpha^{n,i}(\mathbf{x}) = A_{x_i}^n(M^{-i,n}(\mathbf{x})), \quad i = 1, \dots, n, \quad (8)$$

where $A^n : \mathcal{P} \rightarrow \mathcal{A}$ is continuous. Only values on the $(n-1)$ -empirical grid are used by the game; an extension to the whole simplex is assumed in order to formulate uniform regularity and contraction.

Assumption 2.4 (Exact equilibria and uniform empirical regularity). *For every $n \geq 2$, the profile (8) is an exact Markovian Nash equilibrium. Moreover, there is $L_A < \infty$, independent of n , such that*

$$\max_{x \in \mathcal{S}} \|A_x^n(m) - A_x^n(m')\| \leq L_A \|m - m'\|, \quad m, m' \in \mathcal{P}. \quad (9)$$

Define the nonlinear population drift

$$b^n(m) := mA^n(m), \quad m \in \mathcal{P}. \quad (10)$$

Assumption 2.5 (Uniform population contraction). *There is $\kappa > 0$, independent of n , such that*

$$\langle m - m', b^n(m) - b^n(m') \rangle \leq -\kappa \|m - m'\|^2, \quad m, m' \in \mathcal{P}, \quad (11)$$

for every n .

Because every off-diagonal rate is at least $\underline{a}$, the equilibrium chain on $\mathcal{S}^n$ is irreducible. Let Π^n be its unique invariant law, let $\hat{\Pi}^n := \Pi^n \circ (M^n)^{-1}$ be the invariant empirical law, and let ρ_n be the common equilibrium ergodic cost. The generator commutes with coordinate permutations; uniqueness of the invariant law therefore implies that Π^n is exchangeable.

3 Main theorem

Theorem 3.1 (Exact Nash convergence in the contractive stationary class). *Suppose [theorems 2.1](#), [2.4](#) and [2.5](#) hold. Then there is a constant $C < \infty$, independent of n , with the following properties.*

(a) For every n there is a unique $m^n \in \mathcal{P}$ such that

$$b^n(m^n) = m^n A^n(m^n) = 0. \quad (12)$$

For every initial configuration and $t \geq 0$,

$$\mathbb{E} \|M_t^n - m^n\|^2 \leq e^{-2\kappa t} \|M_0^n - m^n\|^2 + \frac{C}{n}. \quad (13)$$

Consequently,

$$\int \|M^n - m^n\|^2 d\Pi^n \leq \frac{C}{n}, \quad W_2^2(\widehat{\Pi}^n, \delta_{m^n}) \leq \frac{C}{n}. \quad (14)$$

(b) Fix a player i , and let that player use an arbitrary admissible Markovian deviation, possibly time-dependent and depending on the full current configuration. If $N_t = M_t^{-i,n}$ denotes the empirical measure of the $n - 1$ nondeviating players, then

$$\mathbb{E} \|N_t - m^n\|^2 \leq e^{-2\kappa t} \|N_0 - m^n\|^2 + \frac{C}{n}, \quad (15)$$

and

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \int_0^T \mathbb{E} \|N_t - m^n\|^2 dt \leq \frac{C}{n}. \quad (16)$$

The same constant works uniformly over all unilateral deviations.

(c) Put $A_n^* := A^n(m^n)$. Then

$$|\rho_n - \mathcal{J}(A_n^*, m^n)| \leq \frac{C}{\sqrt{n}}, \quad (17)$$

and the invariant pair selected by the exact equilibrium has vanishing frozen mean field optimality gap:

$$0 \leq \mathcal{G}(m^n, A_n^*) \leq \frac{C}{\sqrt{n}}. \quad (18)$$

(d) Every subsequential limit of (m^n, A_n^*) belongs to $\mathcal{E}$. In particular,

$$\text{dist}((m^n, A_n^*), \mathcal{E}) \rightarrow 0. \quad (19)$$

(e) Suppose all stationary mean field equilibrium pairs have the same measure $\bar{m}$. Then

$$m^n \rightarrow \bar{m}, \quad \rho_n \rightarrow \mathcal{V}(\bar{m}), \quad W_2(\widehat{\Pi}^n, \delta_{\bar{m}}) \rightarrow 0. \quad (20)$$

For every fixed k , the k -coordinate marginal of Π^n converges to $\bar{m}^{\otimes k}$. If, in addition, the equilibrium generator at $\bar{m}$ is unique, say $(\bar{m}, \bar{A}) \in \mathcal{E}$, then

$$A_n^* \rightarrow \bar{A} \quad (21)$$

and

$$\mathbb{E}_{\Pi^n} \left[\frac{1}{n} \sum_{i=1}^n \|A_{X_i}^n(M^{-i,n}) - \bar{A}_{X_i}\| \right] \rightarrow 0. \quad (22)$$

Remark 3.2 (ε_n -Nash equilibria). If (8) is only an ε_n -Nash equilibrium, then (18) becomes

$$\mathcal{G}(m^n, A_n^*) \leq \varepsilon_n + \frac{C}{\sqrt{n}}.$$

All qualitative convergence conclusions remain valid when $\varepsilon_n \rightarrow 0$.

4 The population fixed point and equilibrium concentration

Lemma 4.1 (Existence and uniqueness of m^n). *For every n , (12) has exactly one solution.*

Proof. For each $m \in \mathcal{P}$, $A^n(m)$ is irreducible and therefore has a unique invariant distribution $\pi(A^n(m))$. The map $m \mapsto \pi(A^n(m))$ is continuous. Brouwer's fixed-point theorem gives $m^n \in \mathcal{P}$ satisfying

$$m^n = \pi(A^n(m^n)),$$

which is equivalent to $m^n A^n(m^n) = 0$.

If m and m' are zeros of b^n , then

$$0 = \langle m - m', b^n(m) - b^n(m') \rangle \leq -\kappa \|m - m'\|^2,$$

so $m = m'$. □

For a configuration $\mathbf{x}$, the drift of its empirical measure under the equilibrium profile is

$$B^n(\mathbf{x}) := \frac{1}{n} \sum_{i=1}^n A_{x_i}^n(M^{-i,n}(\mathbf{x})). \quad (23)$$

The identity

$$M^{-i,n} - M^n = \frac{M^n - \mathbf{e}_{x_i}}{n-1} \quad (24)$$

implies

$$\|M^{-i,n} - M^n\| \leq \frac{\sqrt{2}}{n-1}. \quad (25)$$

Hence, by (9),

$$\|B^n(\mathbf{x}) - b^n(M^n(\mathbf{x}))\| \leq \frac{C}{n}. \quad (26)$$

Lemma 4.2 (Equilibrium Lyapunov estimate). *The bounds (13) and (14) hold.*

Proof. Let

$$V_n(\mathbf{x}) = \|M^n(\mathbf{x}) - m^n\|^2,$$

and let $\mathcal{L}_n$ be the generator of the equilibrium chain. A jump of player i from x_i to y changes M^n by $(\mathbf{e}_y - \mathbf{e}_{x_i})/n$. The exact quadratic expansion gives

$$\mathcal{L}_n V_n(\mathbf{x}) = 2 \langle M^n - m^n, B^n(\mathbf{x}) \rangle \quad (27)$$

$$+ \frac{1}{n^2} \sum_{i=1}^n \sum_{y \neq x_i} A_{x_i y}^n(M^{-i,n}) \|\mathbf{e}_y - \mathbf{e}_{x_i}\|^2. \quad (28)$$

The second line is at most $2(d-1)\bar{a}/n$. Since $b^n(m^n) = 0$, (11) and (26) imply

$$\begin{aligned} 2 \langle M^n - m^n, B^n \rangle &= 2 \langle M^n - m^n, b^n(M^n) - b^n(m^n) \rangle + 2 \langle M^n - m^n, B^n - b^n(M^n) \rangle \\ &\leq -2\kappa \|M^n - m^n\|^2 + \frac{C}{n}, \end{aligned}$$

because the diameter of $\mathcal{P}$ is bounded. Thus

$$\mathcal{L}_n V_n \leq -2\kappa V_n + \frac{C}{n}. \quad (29)$$

Dynkin's formula and Gronwall's inequality give (13), after enlarging C if necessary. Integrating (29) against the invariant law Π^n gives the first bound in (14). The Wasserstein identity

$$W_2^2(\hat{\Pi}^n, \delta_{m^n}) = \int \|m - m^n\|^2 \hat{\Pi}^n(dm)$$

gives the second. $\square$

5 Uniform stability under a unilateral deviation

Fix a player i . That player may use an arbitrary admissible Markovian deviation, while all other players retain their equilibrium feedbacks. Write

$$Y_t := X_t^i, \quad N_t := M_t^{-i,n} = \frac{1}{n-1} \sum_{j \neq i} \mathbf{e}_{X_t^j}.$$

For an opponent $j \neq i$, the empirical measure appearing in that opponent's feedback is

$$M^{-j,n} = N + \frac{\mathbf{e}_Y - \mathbf{e}_{X_j}}{n-1}, \quad \|M^{-j,n} - N\| \leq \frac{\sqrt{2}}{n-1}. \quad (30)$$

The drift of N generated by the opponents' jumps is

$$\tilde{B}^{n,-i} = \frac{1}{n-1} \sum_{j \neq i} A_{X_j}^n(M^{-j,n}). \quad (31)$$

On the other hand,

$$b^n(N) = \frac{1}{n-1} \sum_{j \neq i} A_{X_j}^n(N).$$

Therefore

$$\|\tilde{B}^{n,-i} - b^n(N)\| \leq \frac{C}{n}, \quad (32)$$

uniformly in the deviator's state and strategy.

Lemma 5.1 (One-deviator stability). *The estimates (15) and (16) hold uniformly over all admissible unilateral Markovian deviations.*

Proof. A jump of the deviator does not change N . A jump of one opponent changes N by $(\mathbf{e}_y - \mathbf{e}_x)/(n-1)$. Applying the full, possibly time-dependent, generator to

$$W(N) = \|N - m^n\|^2$$

and using (32) gives, pointwise in time and state,

$$\mathcal{L}_t W \leq -2\kappa W + \frac{C}{n}.$$

The bound contains no term depending on the deviator's chosen rates. Dynkin's formula and Gronwall's inequality give (15). Integrating that estimate over $[0, T]$, dividing by T , and sending $T \rightarrow \infty$ yields (16). $\square$

The lemma is the essential infinite-horizon estimate. A finite-horizon Gronwall bound of order e^{CT}/n would be useless after $T \rightarrow \infty$. Here the perturbation caused by the deviator is dissipated by the same contraction that stabilizes the equilibrium population.

6 From exact Nash optimality to a mean field optimality gap

Set

$$A_n^* := A^n(m^n), \quad c_n^* := \mathcal{J}(A_n^*; m^n).$$

Because $m^n A_n^* = 0$ and A_n^* is irreducible, $m^n = \pi(A_n^*)$ and

$$c_n^* = \sum_x m_x^n c(x, A_x^n(m^n), m^n). \quad (33)$$

Lemma 6.1 (Equilibrium cost freezing). *There is $C < \infty$ such that*

$$|\rho_n - c_n^*| \leq \frac{C}{\sqrt{n}}.$$

Proof. Under Π^n , exchangeability implies

$$\text{Law}_{\Pi^n}(X_i) = \mathbb{E}_{\Pi^n} M^n.$$

By [theorem 4.2](#) and Cauchy–Schwarz,

$$\|\text{Law}_{\Pi^n}(X_i) - m^n\| \leq \mathbb{E}_{\Pi^n} \|M^n - m^n\| \leq \frac{C}{\sqrt{n}}. \quad (34)$$

Also, [\(25\)](#) gives

$$\mathbb{E}_{\Pi^n} \|M^{-i,n} - m^n\| \leq \frac{C}{\sqrt{n}}. \quad (35)$$

The stationary equilibrium cost is

$$\rho_n = \mathbb{E}_{\Pi^n} c(X_i, A_{X_i}^n(M^{-i,n}), M^{-i,n}).$$

Using [theorem 2.1](#), [\(9\)](#), and [\(35\)](#), we may replace $M^{-i,n}$ by m^n in the control and interaction arguments at cost $O(n^{-1/2})$. We then replace the law of X_i by m^n using [\(34\)](#) and boundedness of the running cost. The resulting expression is exactly [\(33\)](#). $\square$

For $\beta \in \mathcal{A}$, let player i deviate to the stationary own-state control whose generator is β , while all opponents retain the equilibrium feedbacks. Let $J_n^{\text{dev}}(\beta)$ denote the deviator’s ergodic cost.

Lemma 6.2 (Stationary deviation cost freezing). *Uniformly over $\beta \in \mathcal{A}$,*

$$|J_n^{\text{dev}}(\beta) - \mathcal{J}(\beta; m^n)| \leq \frac{C}{\sqrt{n}}. \quad (36)$$

Proof. Under this particular deviation, the deviator’s state Y is an autonomous irreducible chain with generator β . Therefore

$$\frac{1}{T} \int_0^T \mathbb{E}[f(Y_t, \beta_{Y_t}) + F(Y_t, m^n)] dt \longrightarrow \mathcal{J}(\beta; m^n).$$

The only difference from the actual deviation cost is the replacement of the opponents’ empirical measure N_t by m^n . By the Lipschitz property of F , Cauchy–Schwarz, and [theorem 5.1](#),

$$\begin{aligned} & \limsup_{T \rightarrow \infty} \frac{1}{T} \int_0^T \mathbb{E} |F(Y_t, N_t) - F(Y_t, m^n)| dt \\ & \leq L_F \left(\limsup_{T \rightarrow \infty} \frac{1}{T} \int_0^T \mathbb{E} \|N_t - m^n\|^2 dt \right)^{1/2} \leq \frac{C}{\sqrt{n}}. \end{aligned}$$

This proves [\(36\)](#). The constant is uniform in β because the stability estimate is uniform and the action space is compact. $\square$

Proposition 6.3 (Frozen optimality gap). *The estimates (17) and (18) hold.*

Proof. The first estimate is [theorem 6.1](#). Since the original profile is an exact Nash equilibrium, it may be tested against every stationary own-state deviation $\beta \in \mathcal{A}$:

$$\rho_n \leq J_n^{\text{dev}}(\beta).$$

Combining [theorems 6.1](#) and [6.2](#) gives

$$\mathcal{J}(A_n^*; m^n) \leq \mathcal{J}(\beta; m^n) + \frac{C}{\sqrt{n}}, \quad \beta \in \mathcal{A}.$$

Taking the minimum over β yields

$$\mathcal{G}(m^n, A_n^*) = \mathcal{J}(A_n^*; m^n) - \mathcal{V}(m^n) \leq \frac{C}{\sqrt{n}}.$$

Nonnegativity follows from the definition of $\mathcal{V}$. □

7 Compactness, equilibrium identification, and chaos

Lemma 7.1 (Continuity). *The maps*

$$A \mapsto \pi(A), \quad (A, m) \mapsto \mathcal{J}(A; m), \quad m \mapsto \mathcal{V}(m), \quad (m, A) \mapsto \mathcal{G}(m, A)$$

are continuous on their respective compact domains. Consequently, $\mathcal{E}$ is compact.

Proof. The invariant distribution of an irreducible finite-state generator depends continuously on its entries. Hence $\mathcal{J}$ is continuous. Continuity of $\mathcal{V}(m) = \min_{A \in \mathcal{A}} \mathcal{J}(A; m)$ follows from compactness of $\mathcal{A}$ and Berge's maximum theorem. The remaining statements are immediate. □

Proposition 7.2 (Identification of all subsequential limits). *Every subsequential limit of (m^n, A_n^*) lies in $\mathcal{E}$, and (19) holds.*

Proof. Take an arbitrary subsequence. Compactness of $\mathcal{P} \times \mathcal{A}$ gives a further subsequence, not relabeled, such that

$$m^n \rightarrow m, \quad A_n^* \rightarrow A.$$

Passing to the limit in $m^n A_n^* = 0$ gives $mA = 0$. By [theorems 6.3](#) and [7.1](#),

$$\mathcal{G}(m, A) = 0.$$

Thus $(m, A) \in \mathcal{E}$. If (19) failed, a subsequence would remain a positive distance from the compact set $\mathcal{E}$; extracting a convergent further subsequence would contradict the conclusion just proved. □

Lemma 7.3 (Exchangeability and empirical convergence imply chaos). *Let Γ^n be exchangeable probability measures on S^n . If the empirical measures M^n converge in probability under Γ^n to a deterministic $m \in \mathcal{P}$, then for every fixed k , the k -coordinate marginal of Γ^n converges to $m^{\otimes k}$.*

Proof. Fix a bounded function $\varphi : S^k \rightarrow \mathbb{R}$. Exchangeability gives

$$\mathbb{E}_{\Gamma^n} \varphi(X_1, \dots, X_k) = \mathbb{E}_{\Gamma^n} \left[\frac{1}{(n)_k} \sum_{i_1, \dots, i_k \text{ distinct}} \varphi(X_{i_1}, \dots, X_{i_k}) \right],$$

where $(n)_k = n(n-1)\cdots(n-k+1)$. The expression inside the expectation is sampling without replacement from the configuration, whereas

$$\int \varphi d(M^n)^{\otimes k} = \frac{1}{n^k} \sum_{i_1, \dots, i_k} \varphi(X_{i_1}, \dots, X_{i_k})$$

is sampling with replacement. Their difference is at most

$$2 \|\varphi\|_\infty \left(1 - \frac{(n)_k}{n^k}\right) \leq \frac{k(k-1)}{n} \|\varphi\|_\infty.$$

Since $m' \mapsto \int \varphi d(m')^{\otimes k}$ is continuous and bounded on $\mathcal{P}$, convergence in probability of M^n to m implies

$$\mathbb{E}_{\Gamma^n} \int \varphi d(M^n)^{\otimes k} \longrightarrow \int \varphi dm^{\otimes k}.$$

The claim follows. $\square$

Proof of theorem 3.1. Parts (a), (b), and (c) are theorems 4.1, 4.2, 5.1 and 6.3. Part (d) is theorem 7.2.

Assume now that all equilibrium pairs have the same measure $\bar{m}$. Every convergent subsequence of m^n has limit $\bar{m}$ by theorem 7.2; compactness therefore implies $m^n \rightarrow \bar{m}$. Moreover,

$$\begin{aligned} W_2^2(\hat{\Pi}^n, \delta_{\bar{m}}) &= \int \|M^n - \bar{m}\|^2 d\Pi^n \\ &\leq 2 \int \|M^n - m^n\|^2 d\Pi^n + 2 \|m^n - \bar{m}\|^2 \longrightarrow 0. \end{aligned}$$

Thus $M^n \rightarrow \bar{m}$ in probability under Π^n , and theorem 7.3 proves convergence of every fixed marginal.

For the values, take an arbitrary subsequence and then a further subsequence along which $A_n^* \rightarrow A$. By theorem 7.2, $(\bar{m}, A) \in \mathcal{E}$, so $\mathcal{J}(A; \bar{m}) = \mathcal{V}(\bar{m})$. Using (17) and continuity of $\mathcal{J}$ gives $\rho_n \rightarrow \mathcal{V}(\bar{m})$ along this further subsequence. Since every subsequence has a further subsequence with the same limit, the full sequence converges.

If the equilibrium generator at $\bar{m}$ is unique, every subsequential limit of A_n^* equals $\bar{A}$, and hence $A_n^* \rightarrow \bar{A}$. Finally, by exchangeability, the triangle inequality, and (9),

$$\begin{aligned} \mathbb{E}_{\Pi^n} \left[\frac{1}{n} \sum_{i=1}^n \|A_{X_i}^n(M^{-i,n}) - \bar{A}_{X_i}\| \right] \\ \leq L_A \mathbb{E}_{\Pi^n} \|M^{-1,n} - m^n\| + \max_x \|A_x^n(m^n) - \bar{A}_x\|. \end{aligned}$$

The first term is $O(n^{-1/2})$ by (14) and (25); the second tends to zero by $A_n^* \rightarrow \bar{A}$. This proves (22). $\square$

8 Quantitative refinements and a checkable contraction condition

8.1 An isolation modulus

Assume for this subsection that $\mathcal{E} = \{(\bar{m}, \bar{A})\}$ is a singleton. Let

$$\mathcal{K} := \{(m, A) \in \mathcal{P} \times \mathcal{A} : mA = 0\}$$

and define

$$\omega(\delta) := \sup \left\{ \|m - \bar{m}\| + \|A - \bar{A}\| : (m, A) \in \mathcal{K}, \mathcal{G}(m, A) \leq \delta \right\}. \quad (37)$$

Compactness, continuity, and uniqueness imply $\omega(\delta) \downarrow 0$ as $\delta \downarrow 0$.

Corollary 8.1 (Isolation-modulus bound). *Under the hypotheses above,*

$$\|m^n - \bar{m}\| + \|A_n^* - \bar{A}\| \leq \omega \left(\frac{C}{\sqrt{n}} \right).$$

If, locally on $\mathcal{K}$,

$$\mathcal{G}(m, A) \geq c_0 (\|m - \bar{m}\| + \|A - \bar{A}\|)^q$$

for some $c_0 > 0$ and $q \geq 1$, then

$$\|m^n - \bar{m}\| + \|A_n^* - \bar{A}\| = O(n^{-1/(2q)}).$$

Proof. The pair (m^n, A_n^*) lies in $\mathcal{K}$, and [theorem 6.3](#) bounds its gap by $C/\sqrt{n}$. The conclusions follow directly from the definition of ω and the assumed error bound. $\square$

8.2 A sufficient weak-interaction condition

Let Q^0 be the complete-graph generator

$$Q_{xy}^0 = \underline{a} \quad (x \neq y), \quad Q_{xx}^0 = -(d-1)\underline{a}. \quad (38)$$

For every zero-sum row vector r ,

$$rQ^0 = -d\underline{a}r. \quad (39)$$

Write $A^n(m) = Q^0 + R^n(m)$. The off-diagonal entries of $R^n(m)$ lie in $[0, \bar{a} - \underline{a}]$, and hence

$$\|R^n(m)\|_{2 \rightarrow 2} \leq \sqrt{\|R^n(m)\|_1 \|R^n(m)\|_\infty} \leq 2(d-1)(\bar{a} - \underline{a}). \quad (40)$$

Proposition 8.2 (A simple contraction criterion). *If*

$$d\underline{a} > 2(d-1)(\bar{a} - \underline{a}) + L_A, \quad (41)$$

then [theorem 2.5](#) holds with

$$\kappa = d\underline{a} - 2(d-1)(\bar{a} - \underline{a}) - L_A > 0.$$

Proof. Let $p, q \in \mathcal{P}$ and $r = p - q$, so $r\mathbf{1} = 0$. Then

$$\begin{aligned} b^n(p) - b^n(q) &= rA^n(p) + q(A^n(p) - A^n(q)) \\ &= rQ^0 + rR^n(p) + q(A^n(p) - A^n(q)). \end{aligned}$$

By (39) and (40),

$$\langle r, rQ^0 + rR^n(p) \rangle \leq [-d\underline{a} + 2(d-1)(\bar{a} - \underline{a})] \|r\|^2.$$

Also, since q is a probability vector,

$$\|q(A^n(p) - A^n(q))\| \leq \max_x \|A_x^n(p) - A_x^n(q)\| \leq L_A \|p - q\|.$$

Taking the inner product with r proves

$$\langle p - q, b^n(p) - b^n(q) \rangle \leq -\kappa \|p - q\|^2.$$

$\square$

Remark 8.3 (Dependence on the off-grid extension). *The finite-player strategy only uses A^n on the $(n-1)$ -empirical grid. This does not create a serious ambiguity under a uniform Lipschitz bound. Any point of $\mathcal{P}$ is within $C_d/(n-1)$ of that grid. Therefore two uniformly L_A -Lipschitz extensions agreeing on the grid differ by $O(1/n)$ uniformly on $\mathcal{P}$. If both induced drifts satisfy the same contraction inequality, their respective fixed points differ by $O(1/n)$ as well. Thus the limiting conclusions are intrinsic to the grid feedbacks rather than to a particular extension.*

9 Relation to the Nash system and scope of the result

Suppose an anonymous exact equilibrium is represented through a finite-player bias function V^n by

$$A_x^n(m) = \gamma^*(x, \Delta_x V^n(\cdot, m)), \quad (42)$$

where the Hamiltonian minimizer γ^* is Lipschitz. A bound of the form

$$\max_x \|\Delta_x V^n(\cdot, m) - \Delta_x V^n(\cdot, m')\| \leq K \|m - m'\|, \quad (43)$$

uniformly in n , would imply (9). Establishing such a bound for exact finite-player Nash bias functions is precisely the kind of Nash-system-to-master-equation regularity bridge that is not proved here.

The present result should therefore be read as follows.

What is proved. For every sequence of symmetric stationary anonymous exact equilibria that already satisfies uniform empirical regularity and uniform population contraction, the invariant empirical measures and values converge to the stationary mean field equilibrium whenever the latter is unique. Even without uniqueness, every cluster point is a stationary mean field equilibrium. The key new implication is the quantitative bridge

$$\text{exact Nash optimality} \implies \mathcal{G}(m^n, A^n(m^n)) = O(n^{-1/2}).$$

What is not proved. The argument does not derive (9) from the Nash system, does not show that (11) follows from the assumptions of [1], does not treat asymmetric or nonanonymous equilibria, and does not treat arbitrary time-dependent exact equilibrium profiles. It also does not prove convergence of the full finite-player bias functions to the master-equation solution U_0 on the entire simplex.

Classification. Accordingly, the theorem is a complete solution of a restricted stationary contractive subproblem, but only *partial progress* on the general open problem in [1, Remark 4.6]. Advertising it as a resolution of that remark without these qualifications would be mathematically misleading.

10 Conclusion

Uniform empirical regularity controls the direct influence of one player on the opponents, while population contraction dissipates that influence over the infinite horizon. Together they yield a deviation-robust Lyapunov estimate and turn exact finite-player Nash inequalities into a vanishing stationary mean field optimality gap. This proves exact-Nash convergence in a meaningful but restricted class. The remaining challenge is structural: obtain the required uniform regularity and stability from the finite-player Nash system itself, or replace them by weaker assumptions that can be verified under the hypotheses of the original ergodic Markovian game.

References

- [1] A. Cohen and E. Zell, Asymptotic Nash equilibria of finite-state ergodic Markovian mean field games, *Mathematics of Operations Research* (2025), DOI: [10.1287/moor.2024.0478](https://doi.org/10.1287/moor.2024.0478); preprint [arXiv:2404.11695](https://arxiv.org/abs/2404.11695).

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- [3] X. Guo and O. Hernández-Lerma, *Continuous-Time Markov Decision Processes: Theory and Applications*, Springer, Berlin, 2009.